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The effect of quinidine on the analgesic effect of codeine

S. H. Sindrup[1], L. Arendt-Nielsen[2], K. Brøsen[1], P. Bjerring[3], H. R. Angelo[4], B. Eriksen[5], and L. F. Gram[1]

[1] Department of Clinical Pharmacology, Odense University, [2] Department of Medical Informatics, Aalborg University, [3] Department of Dermatology, Marselisborg Hospital, Aarhus, [4] Department of Clinical Chemistry, Bispebjerg Hospital, Copenhagen, and [5] DAK-Laboratoriet a/s, Copenhagen, Denmark

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Summary. We have studied the hypoalgesic effect of codeine (100 mg) after blocking the hepatic O-demethylation of codeine to morphine via the sparteine oxygenase (CYP2D6) by quinidine (200 mg). The study was performed in 16 extensive metabolizers of sparteine, using a double-blind, randomized, four-way, cross-over design. The treatments given at 3 h intervals during the four sessions were placebo/placebo, quinidine/placebo, placebo/codeine, and quinidine/codeine. We measured pin-prick pain and pain tolerance thresholds to high energy argon laser stimuli before and 1, 2, and 3 h after codeine or placebo.

After codeine and placebo, the peak plasma concentration of morphine was 6–62 (median 18) $nmol \cdot l^{-1}$. When quinidine pre-treatment was given, no morphine could be detected (< 4 $nmol \cdot l^{-1}$) after codeine. The pin-prick pain thresholds were significantly increased after placebo/codeine, but not after quinidine/codeine compared with placebo/placebo. Both placebo/codeine and quinidine/codeine increased pain tolerance thresholds significantly. Quinidine/codeine and quinidine/placebo did not differ significantly for either pin-prick or tolerance pain thresholds.

These results are compatible with local CYP2D6 mediated formation of morphine in the brain, not being blocked by quinidine. Alternatively, a hypoalgesic effect of quinidine might have confounded the results.

Key words: Codeine, Quinidine; CYP2D6, hypoalgesia, drug interaction

Codeine is eliminated primarily by glucuronidation. O-demethylation to morphine and N-demethylation to norcodeine are minor pathways, each accounting for about 10% of the dose [1] (Fig. 1). Forty years age it was suggested that the hypoalgesic effect of codeine is mediated by its metabolite morphine [2]. This view was supported by subsequent experiments which showed that codeine has much lower affinity for the μ-opioid receptor than morphine (less than 1/3000) [3]. Recently the role of morphine has been questioned by the finding of very low plasma concentrations of morphine after single and repeated doses of codeine [4, 5]. However, the concentrations of the potentially active metabolites of morphine, i. e. normorphine [6] and morphine-6-glucuronide [7], were not taken into consideration. Furthermore, the plasma concentration may not reflect the concentration at the receptor site in the target organ. In fact, it has been suggested that codeine acts through morphine formed locally in the brain [8], but such local transformation has not yet been demonstrated in humans.

Seven per cent of Caucasians are poor metabolizers of sparteine/debrisoquine and the rest are extensive metabolizers [9]. A particular P450, the CYP2D6 (or P450IID6), is the source of this polymorphism, and in the poor metabolizers this isozyme is absent from the liver [10]. It has recently been shown that the O-demethylation of codeine co-segregates with the sparteine/debrisoquine oxidation polymorphism [11–14] and poor metabolizers are practically unable to metabolize codeine to morphine. In a recent study we addressed the role of morphine in codeine hypoalgesia. In 12 extensive and 12 poor metabolizers we found that a single oral dose of 75 mg codeine increased pin-prick pain thresholds to high energy laser stimuli in the extensive but not in the poor metabolizers [15].

Quinidine is a potent inhibitor of the CYP2D6 both in vitro and in vivo [16–18] and the hepatic conversion of codeine to morphine in extensive metabolizers is practically

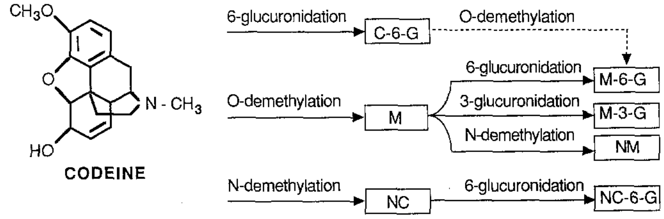

Fig. 1. The pattern of codeine metabolism in humans. Abbreviations: C-6-G = codeine-6-glucuronide, M = morphine, NC = norcodeine, M-6-G = morphine-6-glucuronide, M-3-G = morphine-3-glucuronide, NM = normorphine, and NC-6-G = norcodeine-6-glucuronide

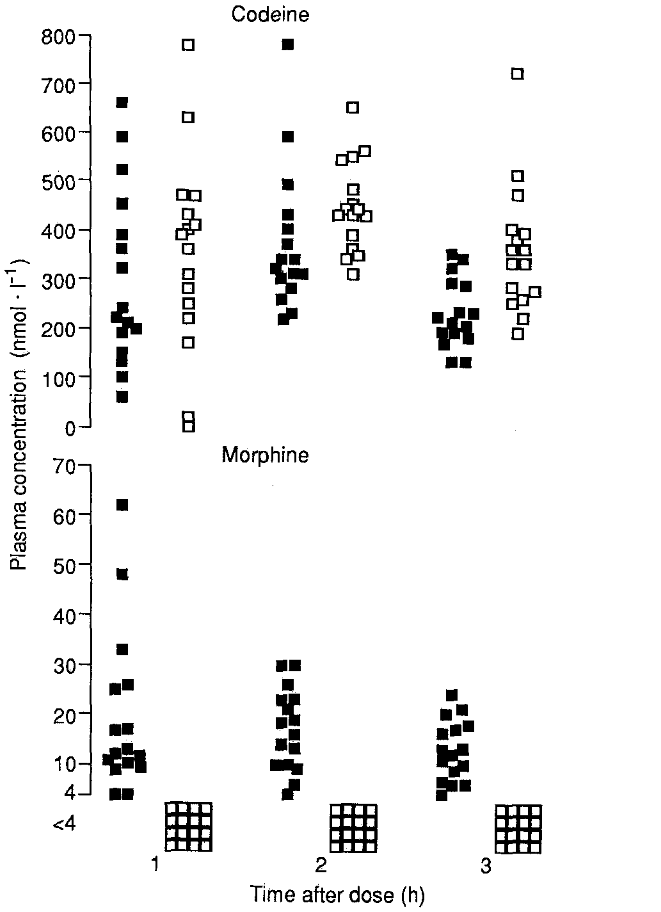

Fig. 2. The plasma concentrations of codeine and morphine 1, 2, and 3 h after 100 mg of codeine in 16 extensive metabolizers of sparteine with (□) and without (■) 200 mg quinidine pre-treatment

abolished after quinidine [19]. We have therefore studied the effect of pre-treatment of 16 healthy extensive metabolizers with 200 mg of quinidine on the hypoalgesia produced by codeine.

Subjects and methods

Subjects

We studied 16 extensive metabolizers of sparteine (12 men and 4 women, aged 21–30 y, median 24) with a metabolic ratio (MR) between sparteine and dehydrosparteines of less than 1 in the 12 h urine [20]. The volunteers were healthy and not regularly taking any drugs (except two of the women who were taking oral contraceptives). Alcohol and additional analgesics were not allowed for 24 h before and after the experiments. The study was approved by the regional ethics committee and the subjects consented to participate on the basis of verbal and written information.

Design and medication

The study was double-blind and randomized. We compared 100 mg of codeine orally with placebo, and both drugs were given with and without pre-treatment with a single oral dose of 200 mg quinidine in a four-way cross-over design. The four different treatment sequence combinations were thus: placebo/placebo, quinidine/placebo, placebo/codeine, and quinidine/codeine. Codeine (25 mg, DAK-Laboratoriet a/s, Copenhagen, Denmark) and codeine-placebo tablets were of identical size and colour, as were the quinidine (200 mg, DAK-Laboratoriet a/s, Copenhagen, Denmark) and quinidine-placebo tablets. Four codeine or codeine-placebo tablets were given 3 h after pre-treatment with one quinidine or quinidine-placebo tablet. The study days were separated by at least two weeks for washout. The pain threshold to laser stimulation was determined immediately before codeine or placebo (i. e. 3 h after quinidine or the corresponding placebo) and then 1, 2, and 3 h after.

Laser stimulation and determination of pain thresholds

A detailed description of the method for determination of pain thresholds to laser stimuli has previously been published [21]. Briefly, the output (wavelenghts 488 and 515 nm) from an argon laser (Model 168, Spectra Physics, USA) was transmitted to the skin through a quartz fibre with a handpiece. The output power was adjustable from 50 mW to approximately 3 W and the energy which reached the skin was measured by an external laser power meter (Ophir, Israël). Laser stimuli of 200 ms duration and a beam diameter of 3 mm were applied to the back (C7 dermatome) of the right hand (pain detection threshold) or left hand (pain tolerance threshold) in a target area of 2×3 cm. After each stimulation, the handpiece was moved to avoid repeated stimulations at identical spots. The pain detection threshold was defined as the stimulus strength which was perceived as a distinct pin-prick; the pain tolerance threshold was defined as the stimulus strength which was perceived as a painful pin-prick followed by an unpleasant burning pain. The thresholds were calculated as the mean of five thresholds determined by ascending series of stimulation.

Plasma drug concentrations

Blood for drug concentration measurements was collected immediately after each pain threshold determination on all study days and stored at $-20\,^{\circ}C$ until analysis. The plasma concentration of codeine was determined by high performance liquid chromatography [22] with a lower limit of detection of 10 $nmol \cdot l^{-1}$. Morphine was assayed by radioimmunoassay (Coat-A-Count, Serum Morphine kit, Diagnostic Product Corporation, Los Angeles, USA) (lower limit of detection 4 $nmol \cdot l^{-1}$; cross-reactivity with codeine, morphine-6-glucoronide, and morphine-3-glucuronide less than 0.2 %).

Data analysis and statistics

Pain thresholds vary both diurnally and from day to day [23]. The thresholds determined on each study day for each individual were therefore transformed (hereafter designated *transformed thresholds*) by multiplication by a factor which standardized the thresholds before codeine or codeine-placebo to a value of 1. The mean of the three transformed post-dose thresholds for each treatment served as a *summary measure* of effect [24]. *A placebo-adjusted treatment response* was calculated for each post dose measurement as the difference between the transformed thresholds on placebo/placebo and on the other treatments.

Statistical analyses were made with Wilcoxon's test for paired observations, the Mann-Whitney test, and Spearman's rank correlation. Furthermore, median and 95 % confidence intervals for differences between treatments were calculated [25].

Results

Morphine was detected in plasma samples from all the volunteers after placebo/codeine, but not after quinidine/codeine (Fig. 2). Codeine concentrations were

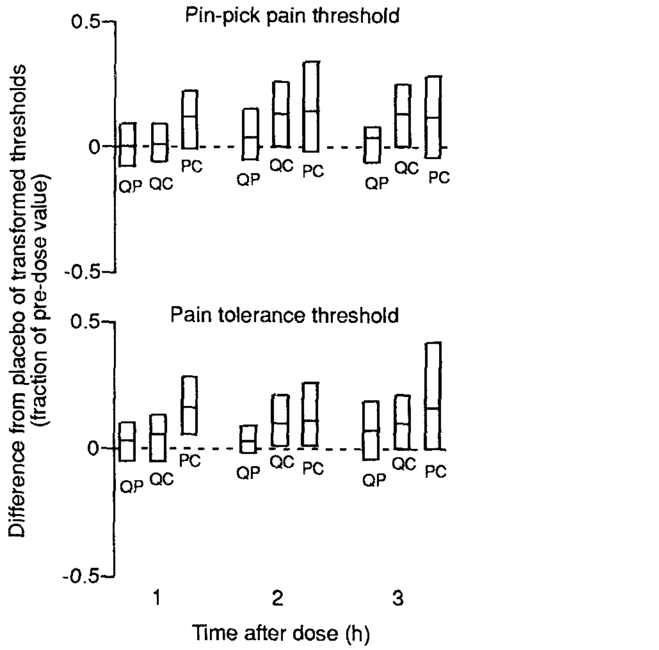

Fig. 3. *Top:* Medians (—) and 95 % confidence intervals (▯) for the differences between placebo/placebo and each of the treatments QP: quinidine/placebo; QC: quinidine/codeine, and PC: placebo/codeine for the pin-prick pain thresholds. The pre-dose thresholds were standardized to a value of 1. *Bottom:* Similar medians and 95 % confidence intervals for the pain tolerance thresholds

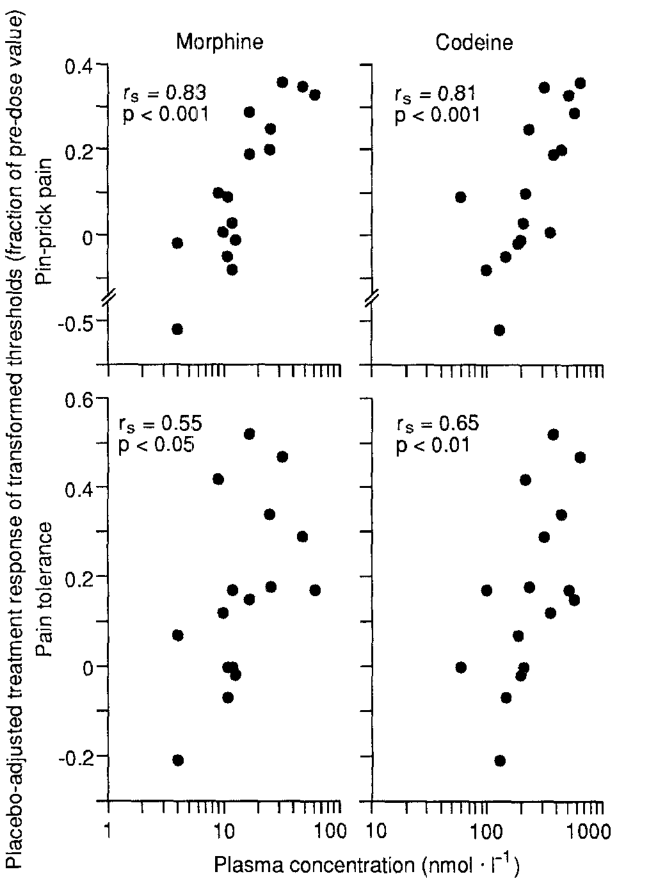

Fig. 4. The effect of 100 mg of codeine 1 h after ingestion in relation to the corresponding plasma concentrations of morphine *(left panel)* and codeine *(right panel)* in each of 16 extensive metabolizers of sparteine. The effect is given as the difference between the transformed thresholds for placebo/codeine and placebo/placebo (placebo adjusted treatment response)

higher after quinidine than after placebo pre-treatment (median 374 vs 264 nmol·l^{-1}, P = 0.003). Neither morphine nor codeine was detected in plasma samples from the volunteers after placebo/placebo or after quinidine/placebo.

The pin-prick pain thresholds measured immediately before codeine or the corresponding placebo on days when quinidine pre-treatment was given (quinidine/placebo and quinidine/codeine) were not significantly different from the thresholds when placebo pre-treatment was given (placebo/placebo and placebo/codeine) (median 1.34 and 1.23 W vs. 1.15 and 1.20 W, P = 0.25–0.74); nor were the corresponding pre-treatment pain tolerance thresholds significantly different (median 1.76 and 1.67 W vs. 1.65 and 1.46 W, P = 0.74–1.0).

Median and 95 % confidence intervals for the differences between placebo/placebo and the other treatments for the transformed pin-prick pain and pain tolerance thresholds are shown in Fig. 3. Significance testing of the summary measure of effect is detailed in Table 1. Pin-prick pain thresholds were only significantly increased by placebo/codeine, whereas pain tolerance thresholds were significantly increased by both placebo/codeine and quinidine/codeine compared with placebo/placebo. Quinidine/codeine did not increase any of the thresholds compared with quinidine/placebo. The most marked differences between the codeine-containing combinations and the placebo-placebo combination were in the tolerance thresholds. Both pin-prick and tolerance thresholds were slightly higher during quinidine/placebo than during placebo/placebo, but the differences did not reach statistical significance.

At 1 h after codeine, the interindividual variation in drug concentrations was most pronounced (Fig. 2), and for the placebo/codeine combination there was a significant correlation between the morphine and codeine concentrations (r_s = 0.67, $P < 0.01$). At this time-point, there was highly significant correlation between the placebo-adjusted treatment response for pin-prick and tolerance thresholds and the plasma concentrations of both morphine (r_s = 0.83, $P < 0.001$ and r_s = 0.55, $P < 0.05$) and co-

Table 1. Significance testing[a] of the effect[b] on pain thresholds of the different treatment combinations[c]

Comparison	P-value	
	Pin-prick pain threshold	Pain tolerance threshold
PC vs. PP	0.06	0.01
QC vs. PP	NS	0.07
QC vs. QP	NS	NS
QC vs. PC	NS	NS
QP vs. PP	NS	NS

[a] Wilcoxon's test
[b] Mean of the three post-dose thresholds transformed to standardize pre-dose thresholds to a value of 1 (summary measure)
[c] Treatment combinations: PC = codeine with placebo pre-treatment, QC = codeine with quinidine pre-treatment, PP = placebo with placebo pre-treatment, QP = placebo with quinidine pre-treatment

Table 2. The number of subjects (out of 16) reporting adverse effects during the treatment combinations[a] before *(A)* and after *(B)* codeine or the corresponding placebo

Side effect	Treatment combination							
	PP		QP		QC		PC	
	A	B	A	B	A	B	A	B
Nausea	0	0	0	0	0	2	0	1
Dry mouth	0	1	1	1	0	2	1	3
Fatigue	2	5	0	4	4	9	3	11
Dizziness	0	0	0	2	0	2	0	4
Headache	0	0	0	1	0	0	0	1

[a] Treatment combinations: PP = placebo with placebo pre-treatment, QP = placebo with quinidine pre-treatment, QC = codeine with quinidine pre-treatment, PC = codeine with placebo pre-treatment

deine (r_s = 0.81, $P < 0.001$ and r_s = 0.65, $P < 0.01$) (Fig. 4). For the 2 and 3 h measurements of both pin-prick pain and pain tolerance thresholds, there was no significant relation to either morphine or codeine concentrations.

The placebo-adjusted treatment responses of the quinidine/codeine combination were not related to the codeine concentrations for either pin-prick pain or pain tolerance at any of the post-dose measurements.

The adverse effects reported on the different treatment combinations are summarized in Table 2. There was no significant difference in the number of adverse effects reported by each subject during any period (P = 0.32).

Discussion

We have found that in 16 extensive metabolizers of sparteine a single oral dose of 100 mg codeine after placebo pre-treatment resulted in plasma morphine concentrations from 6 to 62 $nmol \cdot l^{-1}$ and caused a statistically significant increase in pin-prick pain and pain tolerance thresholds compared with placebo/placebo treatment. We also found that morphine was undetectable in the plasma after a single oral dose of 100 mg codeine preceded by 200 mg quinidine, confirming the importance of CYP2D6 in the O-demethylation of codeine [11–15, 19]. However, we could not unequivocally confirm the role of morphine formation for codeine hypoalgesia [15], since codeine preceded by quinidine also increased pain tolerance thresholds compared with placebo/placebo treatment, but not compared with quinidine/placebo.

There are several possible explanations why codeine may have a hypoalgesic effect in extensive metabolizers treated with quinidine (present study) but not in poor metabolizers (previous study [15]). Although circulating morphine after codeine mainly stems from hepatic O-demethylation, it is possible that codeine hypoalgesia is due to morphine formed locally in the brain [8]. CYP2D6 has been identified in canine brain tissue [26, 27]. Different defects in the CYP2D6 gene [28] explain why the CYP2D6 isozyme is not expressed in the liver [10] or other tissues in the poor metabolizers, and little or no morphine is therefore likely to be formed in the brains of poor metabolizers. Quinidine penetrates the blood-brain barrier poorly [29] and thus may not inhibit CYP2D6 in the brain. This postulated activity of CYP2D6 in brain thus could explain why codeine has a hypoalgesic effect in extensive metabolizers after quinidine pre-treatment despite complete inhibition of hepatic morphine formation.

Another possibilits, supported by the results of animal experiments [30], is that quinidine has an unspecific local analgesic effect and inhibits the release of E-type prostaglandins. Quinidine may therefore have a hypoalgesic effect of its own which is too weak to be detected when it is given alone, but which is detectable when combined with the weak effect of "pure" codeine.

In conclusion, we have confirmed the role of CYP2D6 for the hepatic O-demethylation of codeine to morphine. Our results support the view that metabolically formed morphine is important for codeine analgesia. However, the results of the interaction studies suggest that either the formation of morphine within the CNS or a hypoalgesic effect of quinidine may be a complicating factors.

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Dr. S. H. Sindrup
Department of Clinical Pharmacology
Institute of Medical Biology
Odense University
Winsløwparken 19
DK-5000 Odense C
Denmark